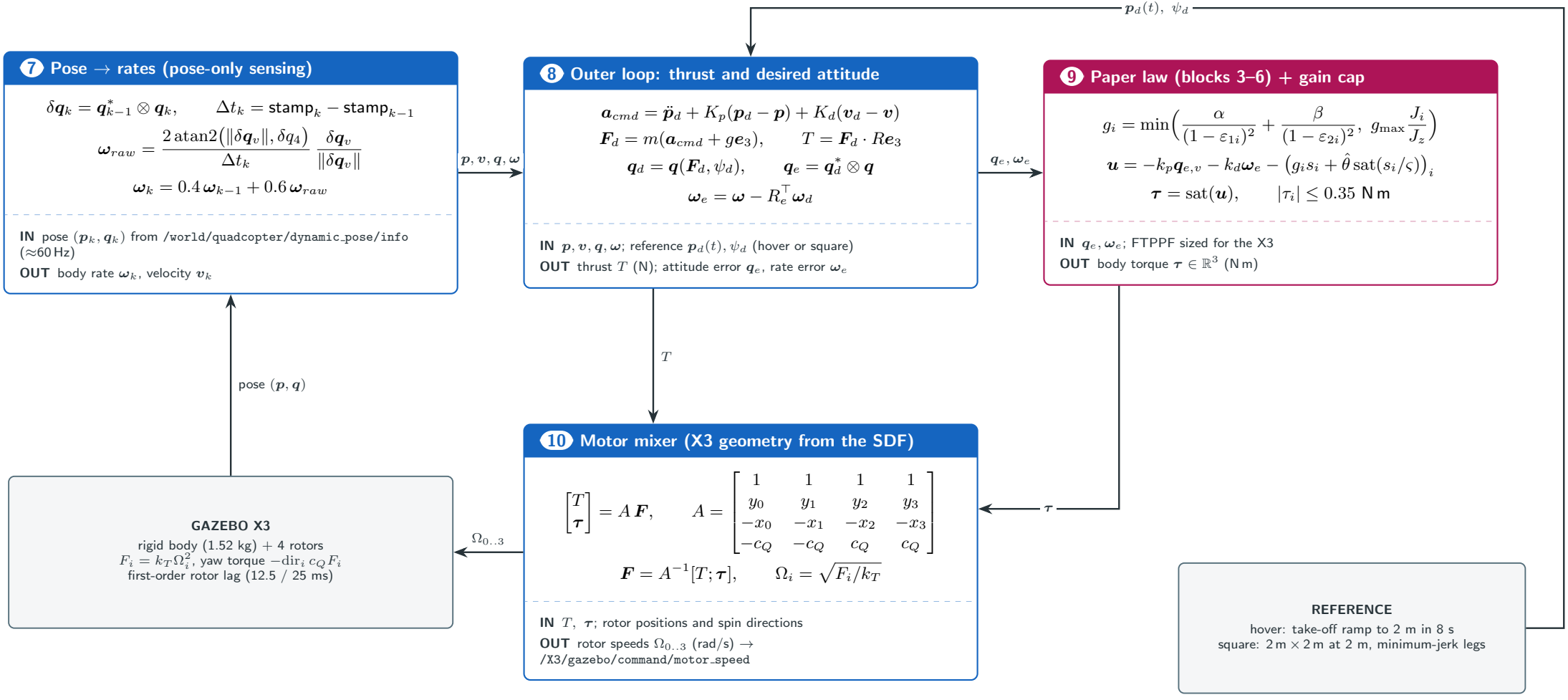

$p_d(t),\ \psi_d$
7 Pose → rates (pose-only sensing)
$\delta q_k = q_{k-1}^* \otimes q_k, \qquad \Delta t_k = \text{stamp}_k - \text{stamp}_{k-1}$
$\omega_{raw} = \dfrac{2\,\text{atan2}\big(\|\delta q_v\|, \delta q_4\big)}{\Delta t_k}\,\dfrac{\delta q_v}{\|\delta q_v\|}$
$\omega_k = 0.4\,\omega_{k-1} + 0.6\,\omega_{raw}$
IN pose (p_k, q_k) from /world/quadcopter/dynamic_pose/info (≈60 Hz)
OUT body rate ω_k, velocity v_k
p, v, q, ω
8 Outer loop: thrust and desired attitude
$a_{cmd} = \ddot{p}_d + K_p(p_d - p) + K_d(v_d - v)$
$F_d = m(a_{cmd} + g e_3), \qquad T = F_d \cdot R e_3$
$q_d = q(F_d, \psi_d), \qquad q_e = q_d^* \otimes q$
$\omega_e = \omega - R_e^\top \omega_d$
IN p, v, q, ω; reference $p_d(t), \psi_d$ (hover or square)
OUT thrust T (N); attitude error q_e, rate error ω_e
q_e, ω_e
9 Paper law (blocks 3–6) + gain cap
$g_i = \min\Big(\dfrac{\alpha}{(1-\varepsilon_{1i})^2} + \dfrac{\beta}{(1-\varepsilon_{2i})^2},\ g_{\max}\dfrac{J_i}{J_z}\Big)$
$u = -k_p q_{e,v} - k_d \omega_e - \big(g_i s_i + \hat{\theta}\,\text{sat}(s_i/\varsigma)\big)_i$
$\tau = \text{sat}(u), \qquad |\tau_i| \le 0.35$ N m
IN q_e, ω_e; FTPPF sized for the X3
OUT body torque $\tau \in \mathbb{R}^3$ (N m)
T
τ
10 Motor mixer (X3 geometry from the SDF)
$\begin{bmatrix} T \\ \tau \end{bmatrix} = A\,F, \qquad A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ y_0 & y_1 & y_2 & y_3 \\ -x_0 & -x_1 & -x_2 & -x_3 \\ -c_Q & -c_Q & c_Q & c_Q \end{bmatrix}$
$F = A^{-1}[T; \tau], \qquad \Omega_i = \sqrt{F_i / k_T}$
IN T, τ; rotor positions and spin directions
OUT rotor speeds $\Omega_{0..3}$ (rad/s) → /X3/gazebo/command/motor_speed
$\Omega_{0..3}$
GAZEBO X3
rigid body (1.52 kg) + 4 rotors
$F_i = k_T \Omega_i^2$, yaw torque $-\text{dir}_i\, c_Q F_i$
first-order rotor lag (12.5 / 25 ms)
pose (p, q)
REFERENCE
hover: take-off ramp to 2 m in 8 s
square: 2 m × 2 m at 2 m, minimum-jerk legs